Exploratory Data Analysis: Comparative Genomic Analysis of Shared Evolutionary Pathways in Diverse Extremophiles

I. Introduction to Data

The breadth of data required for a comparative analysis of extremophiles of a variety of environments is significant, so for the purpose of this exploratory analysis, only thermophile data was prioritized. Thermophiles are historically one of the more heavily studied branches of extremophiles, so this posed an opportunity to actually see how much genomic data is publically available for use in this project. Data for this stage comes from two sources, ThermoBase and NCBI RefSeq. ThermoBase is a centralized database containing environmental and physiological data for thermophile and hyperthermophile species (DiGiacomo et al., 2022). The purpose of this data source in this research is to identify a family of thermophiles that has already been extensively studied. As later analysis in the project will rely on having a large enough set of data to inform statistically significant results, it is important to find large groups of extremophiles that are more closely related, yet also have mesophilic species within the same family. Further explanation will be provided in the analysis section, but using ThermoBase it was decided to move forward searching for thermophile and mesophile genomes from the family Bacillaceae.

NCBI RefSeq functions as the source of all genomic data in the project. RefSeq provides high quality genomic data, along with metadata for each accession (NCBI). This metadata, including genome size, overall gene count, and N50 values is essential for analyzing the quality of genomes. With potentially limited extremophile genomes in general, prioritizing high quality genomes is essential. Upon determining all species would be Bacillaceae using ThermoBase, an NCBI search was conducted, returning 3,323 genomes. The search was limited to complete RefSeq data only, as shown in Figure 1 below.

To ensure the suitability and quality of selected genomes prior to predictive modeling later in the project, this exploratory data analysis investigates three elements: taxonomic structure, genome size and distribution, and sequence quality. In order for later analysis to have meaningful results, the selected genomes need to produce reasonable phylogenies and pangenomes. This is achieved by limiting the selection of species, both thermophilic and mesophilic controls, to the family Bacillaceae. Genome size and distribution of the selected species will show variance, but should be within a reasonable range for optimal pangenomes later. In a similar manner, genome size is also compared with N50 to evaluate quality of the data sequence.

These goals were executed through a python program that determines the family in which most thermophiles in ThermoBase belong. The Bacillaceae thermophiles are then compared to the NCBI search of genomes to find what species have complete sequence data. Mesophile controls are selected based on taxonomic similarity to the selected thermophiles. Upon selection of both thermophile and mesophile Bacillaceae species, metagenomic variables are compared to determine the quality of sequences using a developed R script.

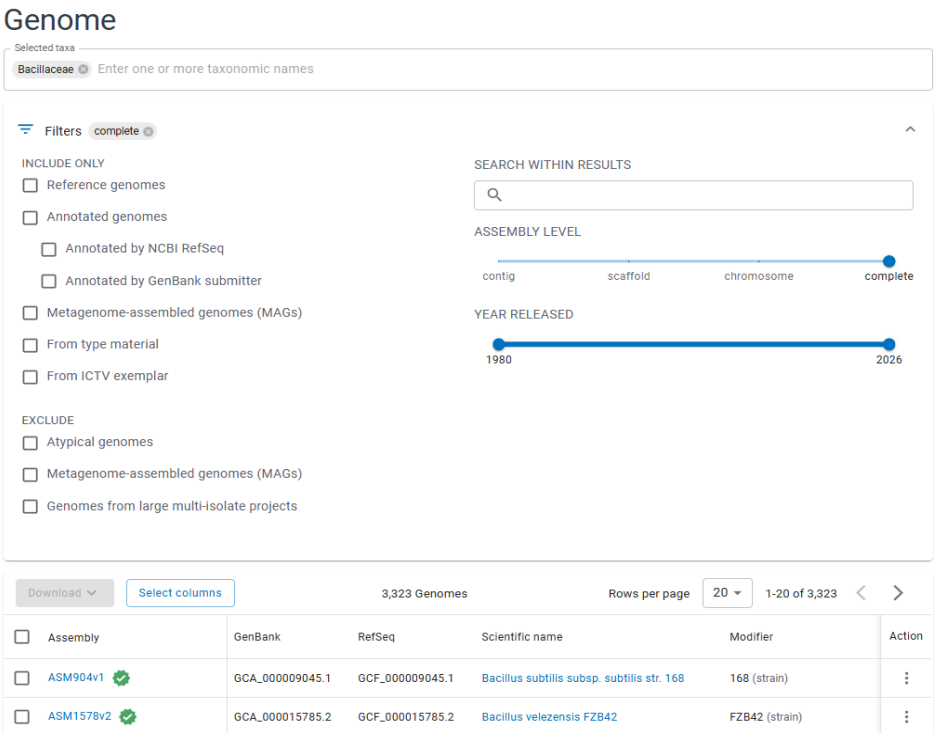


Figure 1: NCBI Genome search interface showing the collection of all complete Bacillaceae genomes (n=3,323)

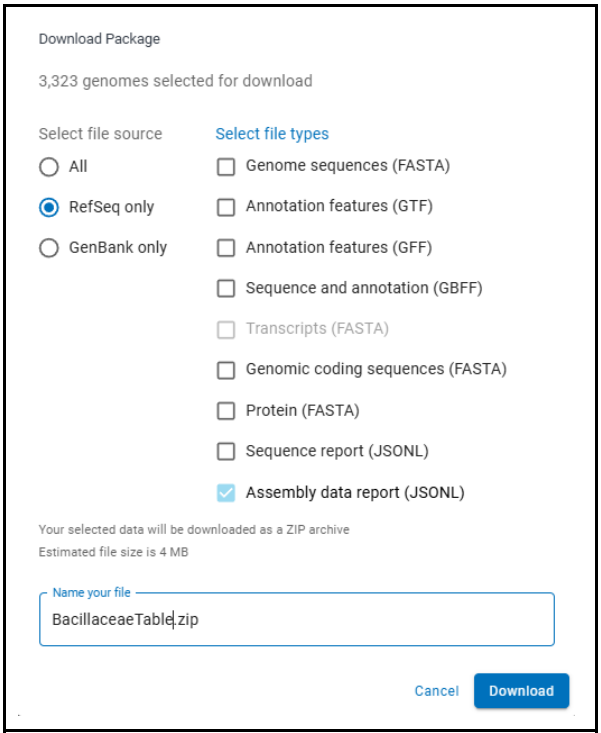


Figure 2: NCBI Genome search download settings to analyze the assembly data reports, only including RefSeq data

II. Exploratory Data Analysis

Use of python in managing species and genome metadata was invaluable. As mentioned above, ThermoBase provided a selection of thermophiles. Though it was already known that Bacillaceae had a number of thermophiles, the created program ThermoCounter (Appendix A) confirmed that Bacillaceae had 93 thermophiles within the database. ThermoCounter exported this list into a .txt, which was then imported into the next program, GenomeSelector.

GenomeSelector (Appendix B) cross-referenced the thermophile list with the imported NCBI RefSeq results. This required flexible code that could work around spelling variation, strain labels, and differing naming conventions. To account for these differences, the script separated organism scientific names into separate "Genus" and "Species" variables. By isolating these taxonomic elements, the program was able to search more flexibly between ThermoBase and NCBI names. Following the matching, duplicate species with more than one genome in NCBI RefSeq were removed, resulting in 9 thermophile genomes being selected (Table 1).

index	Assignment	Organism Scientific Name	Size	Contig N50	Gene Count	Assembly Accession
229	Extremophile	Bacillus licheniformis	4252398	4252398	4431.0	GCF_001596055.1
533	Extremophile	Bacillus halotolerans	4154245	4154245	4286.0	GCF_004006435.1
559	Extremophile	Neobacillus thermocopriae	2843874	2843874	3011.0	GCF_004116995.1
1058	Extremophile	Caldalkalibacillus thermarum TA2.A1	3344914	3344914	3486.0	GCF_019856495.1
1277	Extremophile	Bacillus methanolicus	3560650	3540451	3681.0	GCF_023278615.1
1782	Extremophile	Aeribacillus composti	3834459	3834459	3727.0	GCF_031760945.1
2099	Extremophile	Caldifermentibacillus hisashii	3885383	3885383	3790.0	GCF_039654125.1
2180	Extremophile	Metabacillus halosaccharovorans	5833127	5378041	5734.0	GCF_040959945.1
2803	Extremophile	Salinibacillus aidingensis	3522017	3522017	3599.0	GCF_055393355.1

Table 1: Compiled group of Bacillaceae thermophiles without duplicate species genomes or multiple strains of the same species, n=9.

For every Bacillaceae thermophile, GenomeSelector randomly selected three remaining non-extremophile species of the same genus. This was done to select a control group of mesophile species that have a close evolutionary history with the extremophiles. For adequate machine learning processes later in the project a ratio of 1:3 extremophiles to mesophiles in the same genus was targetted. Where this genus availability was present, the 1:3 was fulfilled. When less than three of the same genus were present the program selected all available controls. Ultimately, GenomeSelector compiled a group of 24 mesophile species (Table 2). All selected species metadata were then consolidated for further statistical analysis in R.

index	Organism Scientific Name	Organism Common Name	Organism Qualifier	Taxonomy id	Assembly Name	Assembly Accession	Source
1471	Bacillus anthracis	anthrax bacterium	strain: BA755	1392	ASM2655977v1	GCF_026559775.1	RefSeq
1780	Bacillus sp. NEB1478	NaN	strain: NEB1478	3073816	ASM3158296v1	GCF_031582965.1	RefSeq
2020	Bacillus sp. FSL R5-0820	NaN	strain: FSL R5-0820	2954560	PS01795	GCF_038593385.1	RefSeq
205	Bacillus thuringiensis	NaN	strain: YWC2-8	1428	ASM142085v1	GCF_001420855.1	RefSeq
832	Bacillus sp. PAMC28571	NaN	strain: PAMC28571	2762327	ASM1425043v1	GCF_014250435.1	RefSeq
518	Bacillus paralicheniformis	NaN	strain: CBMAI 1303	1648923	ASM371102v1	GCF_003711025.1	RefSeq
1760	Neobacillus nitrireducens	NaN	strain: OS1-33	3070683	ASM3091540v1	GCF_030915405.1	RefSeq
2806	Neobacillus sp. Fa239	NaN	strain: Fa239	3433631	ASM5539523v1	GCF_055395235.1	RefSeq
2324	Neobacillus oceani	NaN	strain: SCS-31	3115292	ASM4623759v1	GCF_046237595.1	RefSeq
2809	Caldalkalibacillus mannanilyticus	NaN	strain: JCM 10596	1418	ASM5539756v1	GCF_055397565.1	RefSeq
1166	Bacillus sp. Man122	NaN	strain: Man122	2912825	ASM2216360v1	GCF_022163605.1	RefSeq
2775	Bacillus amyloliquefaciens	NaN	strain: D-1	1390	ASM5481653v1	GCF_054816535.1	RefSeq
326	Bacillus safensis	NaN	strain: BRM1	561879	ASM207721v1	GCF_002077215.1	RefSeq
1930	Aeribacillus sp. FSL K6-8210	NaN	isolate: FSL K6-8210	2954683	PS01971	GCF_037963925.1	RefSeq
1937	Aeribacillus sp. FSL K6-2833	NaN	isolate: FSL K6-2833	2954611	PS01880	GCF_037965815.1	RefSeq
1928	Aeribacillus sp. FSL W8-0870	NaN	isolate: FSL W8-0870	2954706	PS01998	GCF_037963755.1	RefSeq
1305	Cytobacillus firmus	NaN	strain: T8	1399	ASM2361209v1	GCF_023612095.1	RefSeq
1905	Bacillus subtilis	NaN	isolate: FELIX_MS506	1423	ASM3774998v1	GCF_037749985.1	RefSeq
291	Metabacillus harenarius	NaN	strain: Alg07	1547283	ASM188916v1	GCF_001889165.1	RefSeq
3024	Metabacillus sp. KH_069	NaN	strain: KH_069	3465247	ASM5783807v1	GCF_057838075.1	RefSeq
699	Metabacillus sediminilitoris	NaN	strain: DSL-17	2567941	ASM972062v1	GCF_009720625.1	RefSeq
857	Bacillus altitudinis	NaN	strain: BIM B-263	293387	ASM1516089v1	GCF_015160895.1	RefSeq
2794	Halalkalibacter akibai	NaN	strain: JCM 9157	1411	ASM5538202v1	GCF_055382025.1	RefSeq
2270	Oceanobacillus kapialis	NaN	strain: ABH-508	481353	ASM4479018v1	GCF_044790185.1	RefSeq

Table 2: Compiled group of Bacillaceae mesophiles without duplicates, n=24.

After selection of the Bacillaceae species, their metadata were exported into R to explore trends and broad metrics. Table 3 below shows the summary table of metadata among extremophiles and mesophiles. The 9 extremophiles show a mean genome size of 3.91 Mb and a median size of 3.83 Mb, while the mesophiles display a higher mean of 4.49 Mb and median of 4.33. This slight difference is mirrored in the total gene counts, where the extremophile median is 3727 and the mesophile median is 4354.5. Also noteworthy are the N50 values and their similarity to the genome size values. The median N50 for extremophiles is 3.83 and for mesophiles is 4.28. This almost exactly matches the median genome size values. The alignment observed between N50 and genome size proves the sequences are complete genomes as expected, and there are no fragmented sequence gaps. The observed gaps in size and gene count between mesophiles and extremophiles indicate there may be more efficiency within the extremophile genomes. They might have a smaller basic genomic toolkit that is still able to support cell function.

Summary of Selected Bacillaceae Species' Genomic Metadata

Target Cohort	Total Genomes	Distinct Genera	Mean Size (Mb)	Median Size (Mb)	Mean Gene Count	Median Gene Count	Mean N50 (Mb)	Median N50 (Mb)
Extremophile	9	7	3.91	3.83	3971.667	3727.0	3.86	3.83
Mesophile	24	8	4.49	4.33	4519.375	4354.5	4.43	4.28

Table 3: Summary table of all of the species' genomic metadata.

To visualize the variation in genome size across the study groups, a boxplot was created (Figure 3). The figure below shows the average size of extremophile genome are 3.8 Mb, and 4.3 Mb for mesophiles. Unlike the summary table, however, the boxplot allows visualization of the distribution within each group. The graph shows that, were it not for a single extremophile outlier, the average genome size would be even lower. Upon further investigation, the outlier extremophile is Metabacillus halosaccharovorans. It was isolated from the hypersaline lake Aran-Bidgol in Iran, and is actually a halophile (Mehrshad et al., 2013). Further review of ThermoCounter and GenomeSelector is required to determine if a mistake within the pipeline allowed a non-thermophile into the group, or if there was an error within ThermoBase. It could be considered thermotolerant and included in ThermoBase on that basis. Regardless, the distribution of the boxplot highlighted an outlier that, for the purposes of this project, is best to be removed from this dataset.

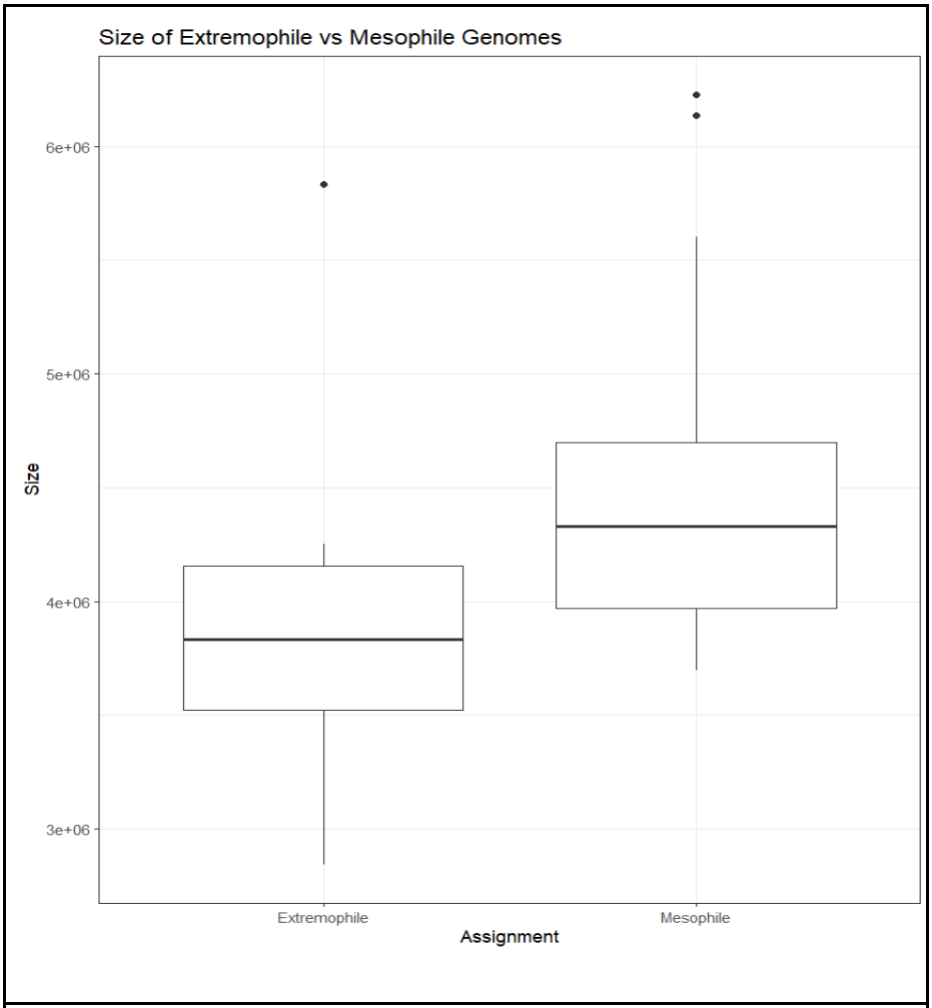


Figure 3: Boxplot showing the genome size comparison in bases between extremophile and mesophile Bacillaceae genomes.

Figure 4 below shows a boxplot of gene count in Bacillaceae extremophiles and mesophiles. It mirrors the boxplot of genome size as expected, and once again the high outlier within the extremophiles is Metabacillus halosaccharovorans.

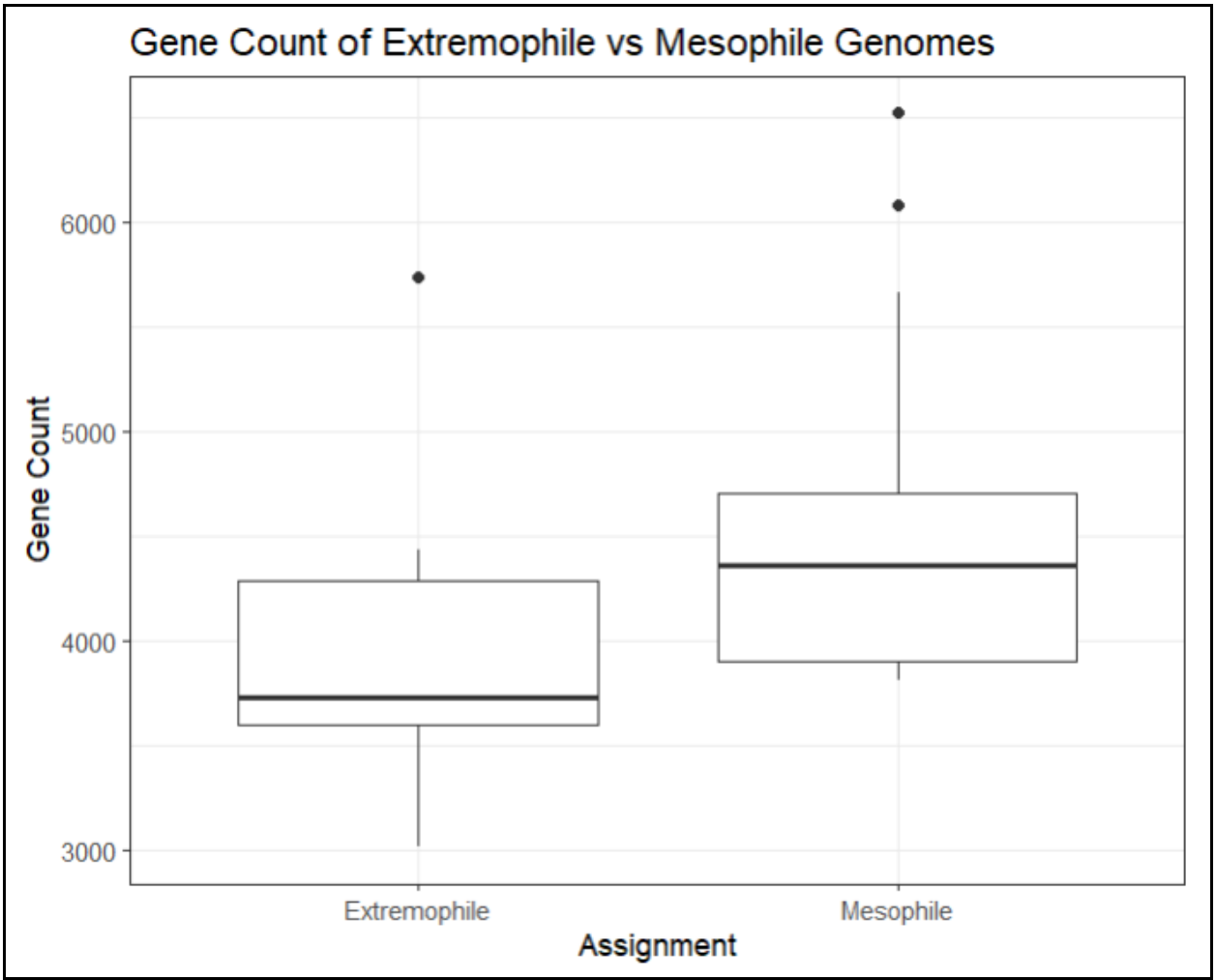


Figure 4: Box plot of gene count between extremophile and mesophile Bacillaceae genomes.

Figure 5 below shows a scatter plot of genome size and gene count, with extremophiles colored red and mesophiles colored green. The relationship between genome size and gene count is shown to be linear, which shows uniform coding density among both groups. Though, there are distinct cluster regions of extremophiles and mesophiles. This is to be expected however, as previously discussed, extremophiles have demonstrated smaller genome size and gene counts. Once again however, Metabacillus halosaccharovorans is the outlier among extremophiles and is located within the mesophile region of the plot.



Figure 5: Scatter plot of gene count by size

Lastly, table 6 below is a contingency table showing the number of extremophiles and mesophiles within each genera represented within the selected species. This table shows where the GenomeSelector program was able to maintain a 1:3 ratio of extremophile to mesophile and where substitutions were made. This visual is useful because it allows for easy understanding of the taxonomic distribution within the selected Bacillaceae species.

Genus	Assignment: Extremophile	Mesophile
aeribacillus	1	3
bacillus	3	11
caldalkalibacillus	1	1
caldifermentibacillus	1	0
cytobacillus	0	1
halalkalibacter	0	1
metabacillus	1	3
neobacillus	1	3
oceanobacillus	0	1
salinibacillus	1	0

Table 6: Contingency table of all genera of selected organisms in the extremophile and mesophile Bacillaceae categories

III. Summary

From this EDA convergence between N50 values and genome sizes showed that selected Bacillaceae species all have complete, non-fragmented genomes. This guarantees that later pangenomes will not have interference due to sequence fragmentation. The summary table, genome size, and count boxplots displayed a noticable difference between thermophile and mesophile groups. Thermophiles exhibited a lower median genome size and count than their mesophile counterparts. This is promising for later analysis, as beyond the obvious phenotypic and environmental differences between these groups, there is a clear difference in genome structure. This difference also supports previous research by Sabath et al in 2013, where it was found that on average with increasing habitat temperature there is a decrease in genome size. In their research they show thermophiles as an example of genome streamlining; a theory that states that natural selection will favor small genomes as a way to minimize energy expendature, increase DNA stability, and lower repair workload (Sabath et al., 2013).

Identifying Metabacillus halosaccharovorans within the thermophile group is a significant takeaway from this EDA. Visualizing trends within each target cohort unexpectedly revealed the inclusion of a halophile within the thermophile group. Further confirmation as to how this occurred is required, but upon brief investigation, Metabacillus halosaccharovorans is not found on the list of Bacillaceae thermophiles created in ThermoCounter. A potential explanation for this problem is the flexible matching within GenomeSelector. Highly flexible matching was necessary to compare between the ThermoBase and NCBI RefSeq names, so it is possible that somehow Metabacillus halosaccharovorans was included in the thermophile group even though it was never in the txt file imported from ThermoCounter. This issue demonstrates the tradeoffs in using data from multiple sources. A flexible matching method is required, or at least a way to uniformly name species between the two data sources. If it is not feasible to make the matching system more rigorous without unduly restricting the extremophile pool, this EDA at

a minimum demonstrates the importance of visualizing the target cohorts as a quality control method.

The EDA also demonstrates that the required genome quality standards for later creating phylogenies, pangenomes and random forest methods are upheld when searching NCBI RefSeq with the previously discussed criteria. In prior project planning it was optimistially stated that a minimum of 15 extremophile genomes per -phile group would be best. While this target is still a fine goal, the ability to only gather 8 complete thermophile genomes shows an adjustment in expectations is required. If a heavily researched group like thermophiles yields only eight complete genomes, less-studied extremophile types are unlikely to provide larger sample sizes. The implications of this finding show that perhaps only the most researched extremophile groups, such as thermophiles, halophiles and acidiphiles, are relevant to the goals of this project.

References

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